

RATCHA BURI AGROMET, Thailand

WMO: 484640

Lat: 13.488N

Lon: 99.796E

Elev: 9

StdP: 101.21

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	17.5	19.0	12.4	9.0	21.2	14.2	10.1	21.9	6.9	26.7	6.3	26.5	1.7	320	0.328

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	9.9	36.6	26.8	35.7	26.6	34.9	26.5	28.2	33.7	27.8	33.0	27.5	32.5	2.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	22.7	30.5	26.6	22.2	30.0	26.2	21.7	29.6	91.0	33.9	89.0	33.0	87.5	32.7	33.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.9	5.0	4.3	DB	15.6	37.9	1.6	1.2	14.4	38.8	13.5	39.5	12.5	40.2	11.3	41.1	(4)
(5)			WB	13.7	30.2	2.2	1.8	12.1	31.5	10.9	32.6	9.6	33.6	8.0	34.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.3	25.8	27.7	29.1	30.4	30.1	29.3	28.7	28.8	28.6	28.0	27.2	25.8
	DBStd	2.05	2.19	1.56	1.77	1.48	1.27	1.01	0.99	0.85	0.96	1.13	1.46	2.21
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6679	491	494	592	613	624	578	580	583	558	559	516	491
	CDD18.3	3637	232	261	333	363	366	328	322	324	308	301	266	232
	CDH23.3	39259	2105	2801	3973	4612	4358	3751	3491	3535	3224	2989	2448	1972
	CDH26.7	16392	809	1273	1937	2406	2054	1551	1334	1365	1188	988	803	684
Wind	WSAvg	1.4	1.5	1.4	1.5	1.5	1.2	1.2	1.2	1.3	1.3	1.2	1.6	1.9
Precipitation	PrecAvg	2415	17	20	53	90	281	375	432	462	387	234	51	15
	PrecMax	2922	102	60	250	359	455	476	630	652	552	366	131	55
	PrecMin	1848	2	1	2	14	98	232	273	302	252	98	0	0
	PrecStd	238	19	17	51	66	77	58	85	81	78	65	37	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	35.2	36.7	38.2	37.5	35.6	34.8	34.9	34.6	33.8	33.7	34.0
		MCWB	24.8	25.2	25.5	26.5	27.7	27.2	26.6	27.0	27.1	26.9	26.6	25.3
	2%	DB	33.2	34.4	35.6	37.1	36.2	34.5	33.7	33.7	33.6	32.8	32.6	32.9
		MCWB	24.6	25.3	25.7	26.9	27.3	27.0	26.3	26.6	26.7	26.8	26.2	25.1
	5%	DB	32.2	33.6	34.8	36.1	35.2	33.5	32.9	32.8	32.7	32.0	31.8	31.7
		MCWB	24.1	25.0	25.8	26.8	27.2	26.7	26.1	26.3	26.5	26.6	26.0	24.6
	10%	DB	31.0	32.7	34.0	35.0	34.0	32.5	32.0	31.9	31.8	31.1	30.7	30.4
		MCWB	23.5	24.6	25.8	26.7	27.0	26.5	26.0	26.1	26.4	26.4	25.6	24.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.4	27.1	27.7	28.7	29.0	28.2	27.6	27.9	28.1	27.9	27.5	26.6
		MCDB	32.0	33.7	33.2	35.1	35.1	33.4	32.4	32.8	32.5	32.1	31.8	31.3
	2%	WB	25.7	26.3	27.1	28.1	28.3	27.7	27.1	27.2	27.4	27.3	27.0	25.9
		MCDB	30.9	32.3	32.9	34.2	33.9	32.5	31.8	31.9	31.8	31.4	31.2	30.9
	5%	WB	25.2	25.9	26.7	27.6	27.7	27.3	26.6	26.9	27.1	27.0	26.6	25.4
		MCDB	29.9	31.5	32.2	33.5	33.0	31.9	31.0	31.4	31.3	30.9	30.5	30.0
	10%	WB	24.6	25.5	26.3	27.2	27.3	26.9	26.3	26.5	26.7	26.7	26.2	24.8
		MCDB	28.9	30.8	31.5	32.8	32.3	31.1	30.3	30.6	30.5	30.1	29.6	29.0
Mean Daily Temperature Range	5% DB	MDBR	10.0	10.7	10.2	9.9	8.6	7.3	7.0	7.2	7.1	6.4	7.0	8.5
		MCDBR	11.4	11.4	11.2	10.9	9.8	8.5	8.3	8.3	8.2	7.6	8.4	10.0
		MCWBR	3.8	3.6	3.3	3.1	2.9	2.7	2.6	2.6	2.8	2.7	2.9	3.3
	5% WB	MCDBR	9.7	10.5	10.5	10.2	9.1	7.8	7.6	7.8	7.7	7.1	7.5	8.8
		MCWBR	3.6	3.6	3.3	3.3	3.1	2.8	2.6	2.8	3.1	2.8	2.9	3.3
Clear-Sky Solar Irradiance	taub	0.539	0.593	0.621	0.552	0.477	0.445	0.441	0.442	0.461	0.572	0.534	0.521	
	taud	1.966	1.863	1.827	2.044	2.306	2.397	2.406	2.388	2.324	1.966	2.045	2.032	
	Ebn at Noon	762	740	730	780	826	844	850	859	849	746	761	766	
	Edh at Noon	180	206	218	175	133	120	119	123	131	184	165	165	
All-Sky Solar Radiation	RadAvg	4.88	5.37	5.53	5.97	5.28	4.85	4.57	4.62	4.75	4.47	4.68	4.70	
	RadStd	0.30	0.25	0.45	0.39	0.47	0.37	0.34	0.35	0.34	0.30	0.35	0.25	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)	N/A	N/A	N/A	N/A	+0.18	+0.20	N/A	N/A	N/A	N/A

Nomenclature: See separate page